CSE 111 Fall 2026: Project Checkpoint 1

YouTutor: Campus AI Learning Platform

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Synopsis

YouTutor is an AI-powered learning platform that helps university students study course material through personalized tutoring and interactive practice. Students enroll in courses, choose topics, ask an AI tutor questions, and complete practice questions with feedback. Their learning sessions are saved so they can review their progress, revisit explanations, and continue studying later.

Students can also browse and learn from sessions shared by other students, then branch off a session to create their own learning path. YouTutor records practice attempts and performance to identify topics a student may need to revisit and suggest targeted practice. The result is a self-paced experience that combines AI tutoring with learning shared across the student community.

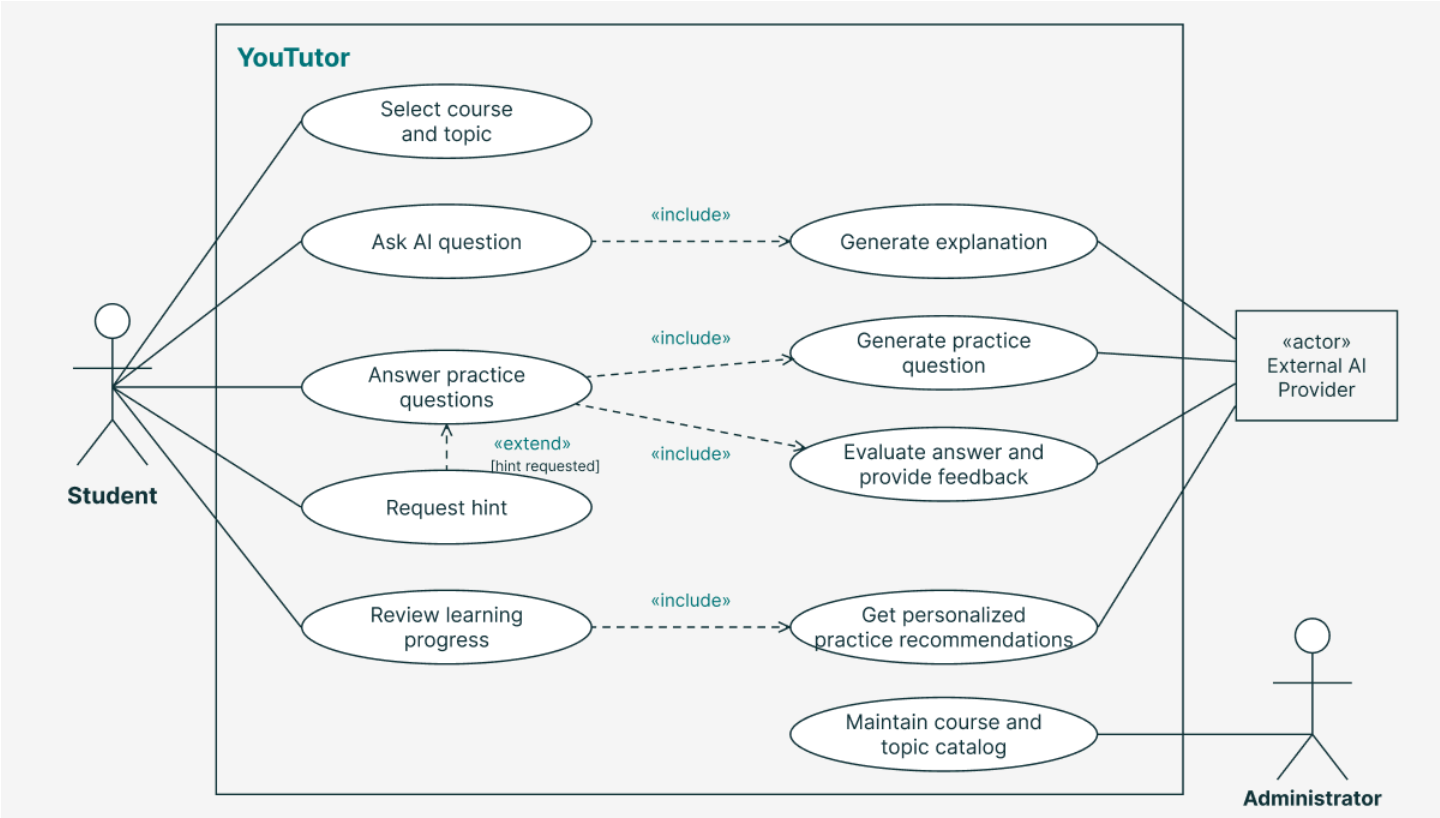
Authors & Contribution: Julian Lee: 50%, Richard Chen: 50%

UML Use Case Diagram

The primary use case is a guided study session. A student selects a course and topic, and the system verifies active enrollment before creating a session. The student asks the AI tutor questions and answers practice questions, the External AI Provider generates explanations, evaluates answers, and provides feedback. The system saves the conversation and each practice attempt. Students can also view sessions shared by others and branch from them to explore a topic in their own way. This makes past study sessions a shared learning resource. Separately, an Administrator maintains the course and topic catalog.

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Use Case Diagram:



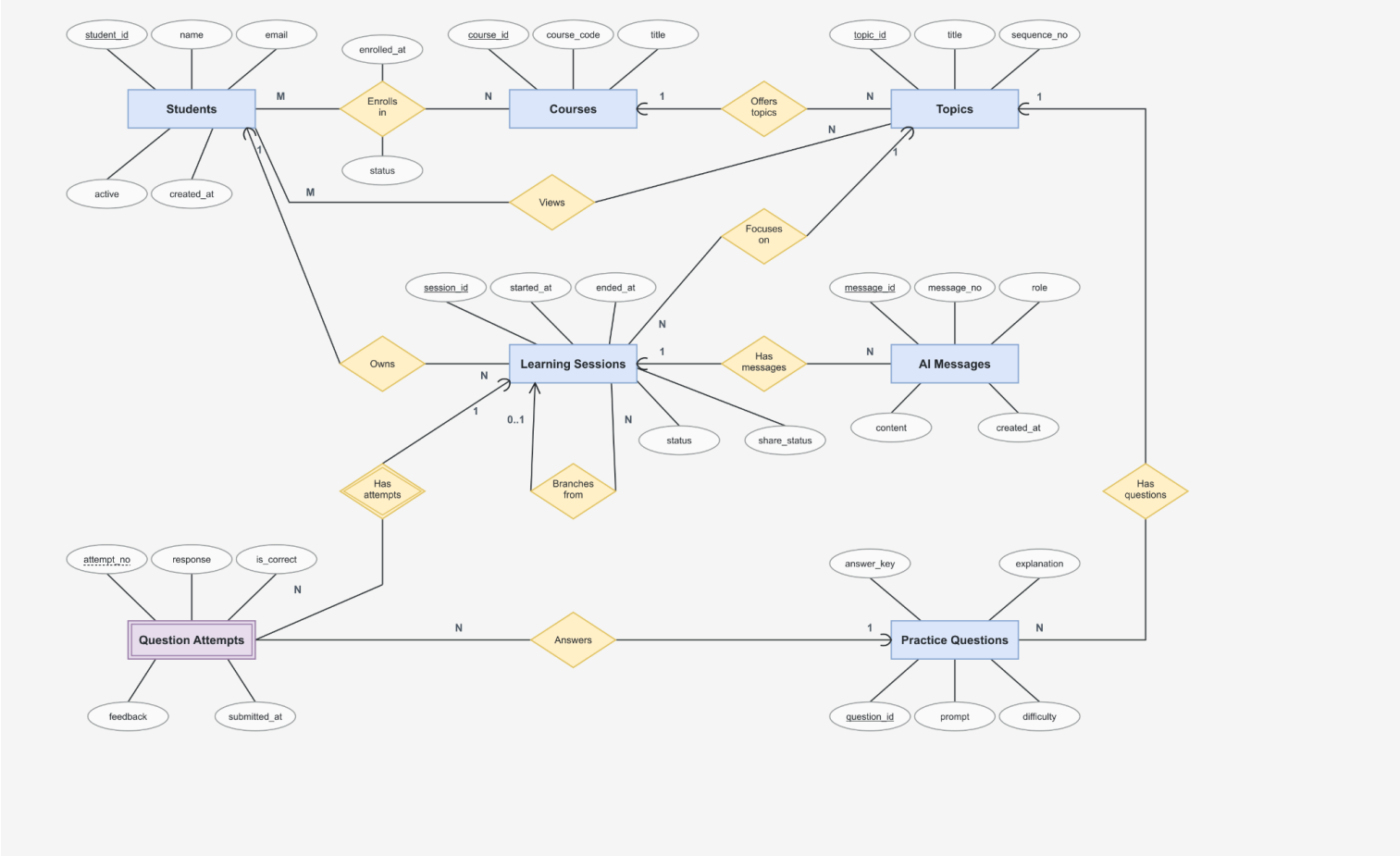
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ER Diagram

The YouTutor database has seven core entities: Students, Courses, Topics, Learning Sessions, AI Messages, Practice Questions, and Question Attempts. Students enroll in Courses and view Topics through two many-to-many relationships, represented by the CourseEnrollments and TopicViews junction tables. Each Course offers multiple Topics.

A Student owns multiple Learning Sessions, each focused on one Topic. A session can be shared, and a new session may branch from an earlier one. Sessions contain ordered AI Messages and Question Attempts. Each Practice Question belongs to a Topic, and each attempt answers one question. Question Attempts are identified by their session and an attempt number, and correctness and feedback are stored on the attempt itself.

ERD-1:



Authors & Contribution: Julian Lee: 50%, Richard Chen: 50%

Relational Schema

The ER design maps to nine relations: six strong entity tables, the weak entity QuestionAttempts, and two junction tables. CourseEnrollments resolves the Students–Courses many-to-many relationship, and TopicViews resolves Students–Topics. QuestionAttempts uses (session_id, attempt_no) as its primary key and references the question answered.

1: Students

```
CREATE TABLE Students (
    student_id INTEGER PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(150) NOT NULL UNIQUE,
    active BOOLEAN NOT NULL DEFAULT TRUE,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);
```

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2: Courses

```
CREATE TABLE Courses (
    course_id INTEGER PRIMARY KEY,
    course_code VARCHAR(30) NOT NULL UNIQUE,
    title VARCHAR(150) NOT NULL
);
```

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3: Topics

```
CREATE TABLE Topics (
    topic_id INTEGER PRIMARY KEY,
    course_id INTEGER NOT NULL,
    title VARCHAR(150) NOT NULL,
    sequence_no INTEGER NOT NULL CHECK (sequence_no > 0),
    UNIQUE (course_id, title),
    UNIQUE (course_id, sequence_no),
    FOREIGN KEY (course_id) REFERENCES Courses(course_id)
```

```
    ON UPDATE CASCADE ON DELETE RESTRICT
);
```

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4: CourseEnrollments

```
CREATE TABLE CourseEnrollments (
    student_id INTEGER NOT NULL,
    course_id INTEGER NOT NULL,
    status VARCHAR(20) NOT NULL DEFAULT 'active'
        CHECK (status IN ('active', 'withdrawn')),
    enrolled_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (student_id, course_id),
    FOREIGN KEY (student_id) REFERENCES Students(student_id)
        ON UPDATE CASCADE ON DELETE CASCADE,
    FOREIGN KEY (course_id) REFERENCES Courses(course_id)
        ON UPDATE CASCADE ON DELETE CASCADE
);
```

Authors & Contribution: Richard Chen: 100%

5. TopicViews

```
CREATE TABLE TopicViews (
    student_id INTEGER NOT NULL,
    topic_id INTEGER NOT NULL,
    PRIMARY KEY (student_id, topic_id),
    FOREIGN KEY (student_id) REFERENCES Students(student_id)
        ON UPDATE CASCADE ON DELETE CASCADE,
    FOREIGN KEY (topic_id) REFERENCES Topics(topic_id)
        ON UPDATE CASCADE ON DELETE CASCADE
);
```

Authors & Contribution: Richard Chen: 100%

6: LearningSessions

```
CREATE TABLE LearningSessions (
    session_id INTEGER PRIMARY KEY,
    student_id INTEGER NOT NULL,
    topic_id INTEGER NOT NULL,
    parent_session_id INTEGER,
    started_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ended_at TIMESTAMP,
    status VARCHAR(20) NOT NULL DEFAULT 'active'
        CHECK (status IN ('active', 'completed')),
    share_status VARCHAR(20) NOT NULL DEFAULT 'private'
        CHECK (share_status IN ('private', 'shared')),
    CHECK (ended_at IS NULL OR ended_at >= started_at),
    CHECK (parent_session_id IS NULL OR parent_session_id <> session_id),
    FOREIGN KEY (student_id) REFERENCES Students(student_id)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    FOREIGN KEY (topic_id) REFERENCES Topics(topic_id)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    FOREIGN KEY (parent_session_id) REFERENCES LearningSessions(session_id)
        ON UPDATE CASCADE ON DELETE RESTRICT
);
```

Authors & Contribution: Julian Lee: 100%

7: AIMessages

```
CREATE TABLE AIMessages (
    message_id INTEGER PRIMARY KEY,
    session_id INTEGER NOT NULL,
    message_no INTEGER NOT NULL CHECK (message_no > 0),
    role VARCHAR(20) NOT NULL
        CHECK (role IN ('student', 'assistant')),
    content TEXT NOT NULL,
```

```
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    UNIQUE (session_id, message_no),
    FOREIGN KEY (session_id) REFERENCES LearningSessions(session_id)
        ON UPDATE CASCADE ON DELETE CASCADE
);
```

Authors & Contribution: Julian Lee: 100%

8: PracticeQuestions

```
CREATE TABLE PracticeQuestions (
    question_id INTEGER PRIMARY KEY,
    topic_id INTEGER NOT NULL,
    prompt TEXT NOT NULL,
    answer_key TEXT NOT NULL,
    explanation TEXT NOT NULL,
    difficulty INTEGER NOT NULL DEFAULT 2
        CHECK (difficulty BETWEEN 1 AND 5),
    FOREIGN KEY (topic_id) REFERENCES Topics(topic_id)
        ON UPDATE CASCADE ON DELETE RESTRICT
);
```

Authors & Contribution: Julian Lee: 100%

9: QuestionAttempts

```
CREATE TABLE QuestionAttempts (
    session_id INTEGER NOT NULL,
    attempt_no INTEGER NOT NULL CHECK (attempt_no > 0),
    question_id INTEGER NOT NULL,
    response TEXT NOT NULL,
    is_correct BOOLEAN NOT NULL,
    feedback TEXT,
    submitted_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (session_id, attempt_no),
```

```
    FOREIGN KEY (session_id) REFERENCES LearningSessions(session_id)
        ON UPDATE CASCADE ON DELETE CASCADE,
    FOREIGN KEY (question_id) REFERENCES PracticeQuestions(question_id)
        ON UPDATE CASCADE ON DELETE RESTRICT
);
```

Authors & Contribution: Richard Chen: 100%